INCH-POUND
MIL-DTL-24643/23F
22 August 2002
SUPERSEDING
MIL-C-24643/23E
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTSU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First -	Copper conductor, uncoated, AWG size 22, class K stranding.
Second -	Extruded silicone rubber insulation (see fourth). Minimum thickness 0.017 inches.
Third -	Insulation jacket of polyamide, per ASTM D 4066, colored (see fourth) or transparent at manufacturer's option. Minimum average wall thickness 0.003 inch. Diameter over insulated and jacketed conductor shall be 0.076 inch minimum. Twisted pair identification code applied by method 6.
Fourth -	Two conductors, one white, one black, twisted together to form a pair having a lay length not greater than 2-1/4 inches. For LSTTSU 1-1/2 only, three conductors, one white, one black, one red, twisted together to form a triad having a lay length not greater than 2-1/4 inches.
Fifth -	The specified number of pairs (see table I) cabled together with a lay no greater than 10 inches. Cabling sequence shall be consecutive, starting with pair No. 1, from the center outward. Fillers shall be used as necessary to form a firm, well rounded assembly.
Sixth -	Binder or combination binder and barrier.
Seventh -	Cross-linked polyolefin jacket. (see table I for thickness).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/23	Size	Overall diameter maximum (inches) LSTTSU 1/	Number of pairs	Jacket thickness min avg. (inch)	Cold bending mandrel diameter maximum (inches)
-01UN	1-1/2	0.330	1-1/2	0.050	5
-02UN	3	0.450	3	0.050	7
-03UN	5	0.540	5	0.050	8
-04UN	10	0.675	10	0.062	9
-05UN	15	0.800	15	0.062	10
-06UN	20	0.870	20	0.062	11
-07UN	30	1.080	30	0.075	13
-08UN	40	1.200	40	0.075	15
-09UN	50	1.400	50	0.075	16
-10UN	60	1.450	60	0.075	17

1/ LSTTSU 1-1/2 comprised of three conductors cabled together to form a triad.

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum 18.60
- Voltage withstand - volts, root mean square, minimum
  - Conductor to conductor 2000
- Insulation resistance - megohms -1000 feet minimum
  - Conductor to conductor 100
  - Conductor continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure
- Watertightness - see MIL-DTL-24643 for limits of water leakage ..... No failure
- Crack resistance ..... No damage

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
- Jacket (When tested at 200°C) 50
- Gas flame 1/ - 1 hour (see note 1) ..... No failure
- Drip - 95 ± 1°C ..... Zero
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35
- Cold bending cable - see Table I for mandrel diameter ..... No damage

Physicals (unaged)

- Insulation
  - Tensile strength - lb/in^2, minimum 750
  - Elongation - percent, minimum 125

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Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation percent, minimum 160

Group C:
Physicals (aged)
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (conductor) - cycles, minimum 25
Permanence of printing (jacket) – cycles, minimum 125
Cable sealant removability ..... No failure

Group D:
Flame propagation (cable) ..... No failure

NOTE: 1. For gas flame test of LSTTSU 1-1/2 the three connectors shall be connected to phases A, B and C of the power supply. For test of all other sizes, connection of conductor pairs shall be in accordance with the following pattern:

(a) Pair #1: One conductor to phase A, one conductor to phase B
(b) Pair #2: One conductor to phase C, one conductor to phase A
(c) Pair #3: One conductor to phase B, one conductor to phase C
(d) Pair #4: One conductor to phase A, one conductor to phase B
(e) Pair #5: One conductor to phase C, one conductor to phase A
(f) Pair #6 through Pair #60: In similar sequence.

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:
Cold working (minus 20 ± 2°C) ..... No damage
Gas flame (3 hours) (see note 1) ..... No failure
Specific gravity of extruded silicone insulation, maximum 1.55
Aging and compatibility (cable) (125 ± 5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18

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Halogen content -percent, maximum
- Jacket 0.2
- Fillers 0.2

Immersion (jacket)
- Tensile strength - percent of unaged, minimum 50
- Elongation - percent of unaged, minimum 50

Smoke index, maximum
- Jacket 25
- Fillers 45
- Insulation 35

Toxicity index, maximum
- Jacket 5
- Fillers 5
- Insulation 1.5

Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure

UNIT ORDERING LENGTHS:

Type and Size	Feet (nominal)
LSTTSU -1-1/2 through -10	2500
LSTTSU -15 and -20	1500
LSTTSU -30 through -60	1000

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-021)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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